

			Optimistic	Most Likely	Pessimistic	Estimate		
Requirement Analysis	Area	Details						
	Business requirement	Manage chicken, cattle, dairy, fish	1	2	3	2.00		
	User requirement	Farm manager, vet, accountant, store officer	1	2	3	2.00		
	Functional requirement	Feed entry, milk record, vaccination, fish harvest	2	3	5	3.17		
	Reporting requirement	Profit/loss, mortality, production, inventory	2	3	5	3.17		
	Compliance requirement	Health, biosecurity, audit trail	0.25	0.5	1	0.54	10.88	
Feasibility Study	Feasibility	Meaning				0.00		
	Technical	Can the system be built with available technology?	1	2	3	2.00		
		Can farm staff use it easily?	0.25	0.5	1	0.54		
	Financial	Is the cost justified by business benefit?	0.25	0.5	1	0.54		
	Schedule	Can it be completed within target time?	0.25	0.5	1	0.54		
	Legal/compliance	Does it follow local rules and audit needs?	0.25	0.5	1	0.54	4.17	
System Design	Design Component	Example				0.00		
	Architecture	Web app, mobile app, cloud/server, failover	1	2	3	2.00		
	Database design	Animal, flock, pond, feed, sales tables	2	3	5	3.17		
	UI/UX design	Simple forms for daily farm entry	3	5	8	5.17		
	Workflow design	Purchase → stock receive → feed issue	3	5	8	5.17		
	API Design		3	5	8	5.17		
	Security design	Role-based access control	2	3	5	3.17		
	Integration design	SMS, accounting, payment, IoT sensors	2	3	5	3.17	27.00	
						0.00		
Development / Coding	Development Environment Rreadiness		2	3	5	3.17		
	Development					128.70	131.87	
Testing	Testing Type	Purpose				0.00		
	Test Environment Readiness		1	2	3	2.00		
	Unit testing	Test each function separately	2	3	5	3.17		
	Integration testing	Check modules working together	2	3	5	3.17		
	System testing	Test full system end-to-end	2	3	5	3.17		
	UAT	User Acceptance Testing by farm users	2	3	5	3.17		
	Security testing	Check login, access control, data protection	1	2	3	2.00		
	Performance testing	Check speed with many records	2	3	5	3.17	19.83	
Training and Transition		Purpose						
		Familiarizing rural workers with handling digital systems efficiently	3	5	8	5.1666667	5.16666666666667	
Deployment / Implementation	Step	Example				0.00		
	Server/cloud setup		2	3	5	3.17		
	Data migration	Existing animal, stock, customer data	2	3	5	3.17		
	User training	Farm manager, store officer, accountant	1	2	3	2.00		
	Pilot run	Start with one farm/unit first	2	3	5	3.17		
	Go-live	Full system launch	1	2	3	2.00	13.50	
						212.41		
						Mandays Rate	7,500	1,593,063
						5% Contengency		1,672,716

Module	Features and Functions	Description	Optimistic	Most Likely	Pessimistic	Estimate
Master Data Setup	Farm profile	Farm name, location, owner, branches	0.5	1	2	1.083333
	Shed/pond/barn setup	Poultry shed, cattle barn, dairy unit, fish pond	0.5	1	2	1.083333
	Breed setup	Broiler, layer, cow breed, fish species	0.5	1	2	1.083333
	Employee setup	Farm manager, vet, feeding staff, workers	0.5	1	2	1.083333
	Supplier/customer setup	Feed supplier, medicine supplier, milk buyer, fish buyer	0.5	1	2	1.083333
	Supplier/customer setup	Feed supplier, medicine supplier, milk buyer, fish buyer	0.5	1	2	1.083333
Livestock / Stock Management	Flock registration	Batch number, breed, arrival date, quantity	0.5	1	2	1.083333
	Mortality tracking	Daily death count and reason	0.5	1	2	1.083333
	Feed consumption	Feed type, quantity, cost	0.5	1	2	1.083333
	Vaccination schedule	Vaccine name, date, dosage	0.5	1	2	1.083333
	Egg production	Daily egg count, broken eggs, saleable eggs	0.5	1	2	1.083333
	Weight monitoring	Average bird weight by age	0.5	1	2	1.083333
Cattle Management	Sales tracking	Bird sale, egg sale, batch closing	0.5	1	2	1.083333
	Animal registration	Tag number, breed, age, sex, purchase date	0.5	1	2	1.083333
	Weight tracking	Regular weight record	0.5	1	2	1.083333
	Feed plan	Dry feed, green grass, concentrate	0.5	1	2	1.083333
	Health record	Disease, treatment, medicine	0.5	1	2	1.083333
	Vaccination/deworming	Schedule and history	0.5	1	2	1.083333
Dairy Management	Breeding record	Heat, insemination, pregnancy check	0.5	1	2	1.083333
	Sale/purchase record	Animal buying and selling	0.5	1	2	1.083333
	Lactation tracking	Cow-wise lactation cycle	0.5	1	2	1.083333
	Milk production	Morning/evening milk record	0.5	1	2	1.083333
	Milk quality	Fat %, SNF, rejected milk	0.5	1	2	1.083333
	Calving record	Birth date, calf details	0.5	1	2	1.083333
Fish/ Aquaculture Management	Breeding management	AI/service date, pregnancy status	0.5	1	2	1.083333
	Feed vs milk analysis	Cost of feed compared with milk output	0.5	1	2	1.083333
	Milk sales	Customer-wise milk delivery and billing	0.5	1	2	1.083333
	Pond registration	Pond size, water capacity, location	0.5	1	2	1.083333
	Fish stocking	Species, quantity, stocking date	0.5	1	2	1.083333
	Feed management	Feed type, quantity, feeding schedule	0.5	1	2	1.083333
Feed Management	Water quality	pH, oxygen, temperature, ammonia	0.5	1	2	1.083333
	Mortality tracking	Dead fish count and reason	0.5	1	2	1.083333
	Growth monitoring	Sampling weight, average size	0.5	1	2	1.083333
	Harvesting	Partial/full harvest record	0.5	1	2	1.083333
	Fish sales	Buyer, quantity, rate, revenue	0.5	1	2	1.083333
	Feed purchase	Supplier, quantity, rate	0.5	1	2	1.083333
Health, Medicine & Vaccination Management	Feed stock	Opening, received, issued, balance	0.5	1	2	1.083333
	Feed formulation	Mix ingredients for poultry/cattle/fish	0.5	1	2	1.083333
	Feed issue	Shed-wise, animal-wise, pond-wise	0.5	1	2	1.083333
	Feed conversion ratio	FCR for poultry and fish	0.5	1	2	1.083333
	Feed cost analysis	Cost per kg meat, milk, egg, or fish	0.5	1	2	1.083333
	Disease record	Animal/flock/pond-wise disease	0.5	1	2	1.083333
Production Management	Treatment record	Medicine, dosage, vet name	0.5	1	2	1.083333
	Vaccination schedule	Planned and completed vaccination	0.5	1	2	1.083333
	Deworming schedule	Cattle/dairy specific	0.5	1	2	1.083333
	Medicine stock	Purchase, issue, expiry date	0.5	1	2	1.083333
	Vet visit record	Diagnosis and recommendation	0.5	1	2	1.083333
	Poultry	Eggs, live bird weight, meat production	0.5	1	2	1.083333
Inventory & Store Management	Dairy	Milk production	0.5	1	2	1.083333
	Cattle	Weight gain, calf birth, meat output	0.5	1	2	1.083333
	Fish	Fish growth, harvest quantity	0.5	1	2	1.083333
	Feed stock	Poultry feed, cattle feed, fish feed	0.5	1	2	1.083333
	Medicine	Vaccine, antibiotic, vitamins	0.5	1	2	1.083333
	Farm materials	Trays, nets, buckets, tools	0.5	1	2	1.083333
Sales & Distribution Management	Equipment	Milking machine, generator, water pump	0.5	1	2	1.083333
	Packaging	Egg tray, milk container, fish box	0.5	1	2	1.083333
	Egg sales	Retail/wholesale egg sales	0.5	1	2	1.083333
	Milk sales	Daily milk supply	0.5	1	2	1.083333
	Cattle sales	Live animal sales	0.5	1	2	1.083333
	Fish sales	Harvest sale	0.5	1	2	1.083333
Procurement Management	Poultry sales	Broiler/layer sale	0.5	1	2	1.083333
	Animal purchase	Chicks, cows, calves, fingerlings	0.5	1	2	1.083333
	Feed purchase	Poultry, cattle, fish feed	0.5	1	2	1.083333
	Medicine purchase	Vaccine, treatment medicine	0.5	1	2	1.083333
	Equipment purchase	Farm tools and machinery	0.5	1	2	1.083333
	Land purchase/lease		0.5	1	2	1.083333
Finance & Accounting	Income tracking	Egg, milk, cattle, fish, chicken sales	0.5	1	2	1.083333
	Expense tracking	Feed, medicine, labor, electricity	0.5	1	2	1.083333
	Cash/bank book	Transaction tracking	0.5	1	2	1.083333
	Accounts payable	Supplier dues	0.5	1	2	1.083333
	Accounts receivable	Customer dues	0.5	1	2	1.083333
	Profit & loss	Farm-wise profitability	0.5	1	2	1.083333
Asset & Maintenance Management	Cost center	Poultry, dairy, cattle, fish unit-wise cost	0.5	1	2	1.083333
	Machinery	Milking machine, pump, generator	0.5	1	2	1.083333
	Vehicles	Delivery van, pickup	0.5	1	2	1.083333
	Farm equipment	Weighing scale, incubator, oxygen pump	0.5	1	2	1.083333
	Infrastructure	Shed, pond, barn, cold storage	0.5	1	2	1.083333
	Poultry	Mortality %, FCR, egg production %, feed cost	0.25	0.5	1	0.541667
Analytics & Dashboard	Dairy	Milk per cow, lactation performance, fat %, feed cost/liter	0.25	0.5	1	0.541667
	Cattle	Weight gain, breeding success, health cost	0.25	0.5	1	0.541667
	Fish	Survival rate, FCR, harvest quantity, water quality	0.25	0.5	1	0.541667
	Finance	Revenue, expense, profit, cost per unit	0.25	0.5	1	0.541667
	Inventory	Feed stock, medicine expiry, reorder alerts	0.25	0.5	1	0.541667
	Asset register	Asset details and value	0.25	0.5	1	0.541667
Mobile App / Field Data Entry	Maintenance schedule	Preventive maintenance	0.25	0.5	1	0.541667
	Repair record	Cost and vendor	0.25	0.5	1	0.541667
	Depreciation	Accounting value reduction	0.25	0.5	1	0.541667
	Purchase requisition	Department-wise request	0.25	0.5	1	0.541667
	Purchase order	Supplier order	0.25	0.5	1	0.541667
	Supplier quotation	Price comparison	0.25	0.5	1	0.541667
Reports Module	Goods receive	Quantity and quality check	0.25	0.5	1	0.541667
	Supplier payment	Payable tracking	0.25	0.5	1	0.541667
	Sales order	Quantity, rate, delivery date	0.25	0.5	1	0.541667
	Invoice	Bill generation	0.25	0.5	1	0.541667
	Payment collection	Cash, bank, due	0.25	0.5	1	0.541667
	Delivery tracking	Product dispatch	0.25	0.5	1	0.541667
	Customer ledger	Outstanding balance	0.25	0.5	1	0.541667
	Daily production report	Overall daily output	0.25	0.5	1	0.541667
	Unit-wise productivity	Shed/barn/pond performance	0.25	0.5	1	0.541667
	Loss report	Mortality, rejected milk, broken eggs	0.25	0.5	1	0.541667
	Production trend	Weekly/monthly comparison	0.25	0.5	1	0.541667
	Daily entry	Feed, mortality, milk, egg, water data	2	3	5	3.166667
	Offline mode	Works without internet	2	3	5	3.166667
	Photo upload	Disease, pond, animal photo	2	3	5	3.166667
	Alerts	Vaccination, low stock, disease warning	2	3	5	3.166667
	QR/tag scan	Animal or batch identification	2	3	5	3.166667
	Poultry reports	Flock performance, mortality, egg production	0.25	0.5	1	0.541667
	Dairy reports	Cow-wise milk report, lactation report	0.25	0.5	1	0.541667
	Cattle reports	Animal health, breeding, weight gain	0.25	0.5	1	0.541667
	Fish reports	Pond performance, water quality, harvest	0.25	0.5	1	0.541667
	Inventory reports	Feed stock, medicine stock, expiry	0.25	0.5	1	0.541667
	Financial reports	P&L, cash flow, customer/supplier ledger	0.25	0.5	1	0.541667
	HR reports	Attendance, salary, duty roster	0.25	0.5	1	0.541667
						116.0417
					Round	117
					Buffer 10%	128.7

Capital Expenditure	Amount	
	₺1,672,716	
Operational Expenditure	Amount (Monthly)	
	Amount (Yearly)	
	Server Rent	₺5,000.00
	Annual Maintenance Contract (AMC)	₺334,543
Cost Analysis	First Year	
	Upcoming Years	
	Total Expenditure	₺1,732,716
		₺394,543
Yearly Revenue	₺492,000.00	
Yearly Benefit	₺97,456.87	
Cost vs Benefit	First Year	
	Upcoming Years	
	Yearly Benefit	₺97,457
		-₺1,240,716
Payback Period	years	months
	17.16	206.0